

Table 9-1: Turtle Functions in Python's *turtle* Module and JavaScript's *jtg* Library

Python	JavaScript	Description
<code>goto(x, y)</code>	<code>xy(x, y)</code>	Moves the turtle to the x, y coordinates.
<code>setheading(deg)</code>	<code>heading(deg)</code>	Sets the turtle's heading. In Python, 0 degrees is east (right). In JavaScript, 0 degrees is north (up).
<code>forward(steps)</code>	<code>fd(steps)</code>	Moves the turtle a number of steps forward in the heading it is facing.
<code>backward(steps)</code>	<code>bk(steps)</code>	Moves the turtle a number of steps in the heading opposite from the one it is facing.
<code>left(deg)</code>	<code>lt(deg)</code>	Turns the turtle's heading to the left.
<code>right(deg)</code>	<code>rt(deg)</code>	Turns the turtle's heading to the right.
<code>penup()</code>	<code>pu()</code>	"Raises the pen" so that the turtle stops drawing as it moves.
<code>pendown()</code>	<code>pd()</code>	"Lowers the pen" so that the turtle starts drawing as it moves.
<code>pensize(size)</code>	<code>thickness(size)</code>	Changes the thickness of the lines the turtle draws. The default is 1.
<code>pencolor(color)</code>	<code>color(color)</code>	Changes the color of the lines the turtle draws. This can be a string of a common color such as red or white. The default is black.
<code>xcor()</code>	<code>get.x()</code>	Returns the turtle's current x position.
<code>ycor()</code>	<code>get.y()</code>	Returns the turtle's current y position.
<code>heading()</code>	<code>get.heading()</code>	Returns the turtle's current heading as a floating-point number from 0 to 359. In Python, 0 degrees is east (right). In JavaScript, 0 degrees is north (up).
<code>reset()</code>	<code>reset()</code>	Clears any drawn lines, and moves the turtle back to the original position and heading.
<code>clear()</code>	<code>clean()</code>	Clears any drawn lines but doesn't move the turtle.

The functions listed in [Table 9-2](#) are available only in the Python `turtle` module.